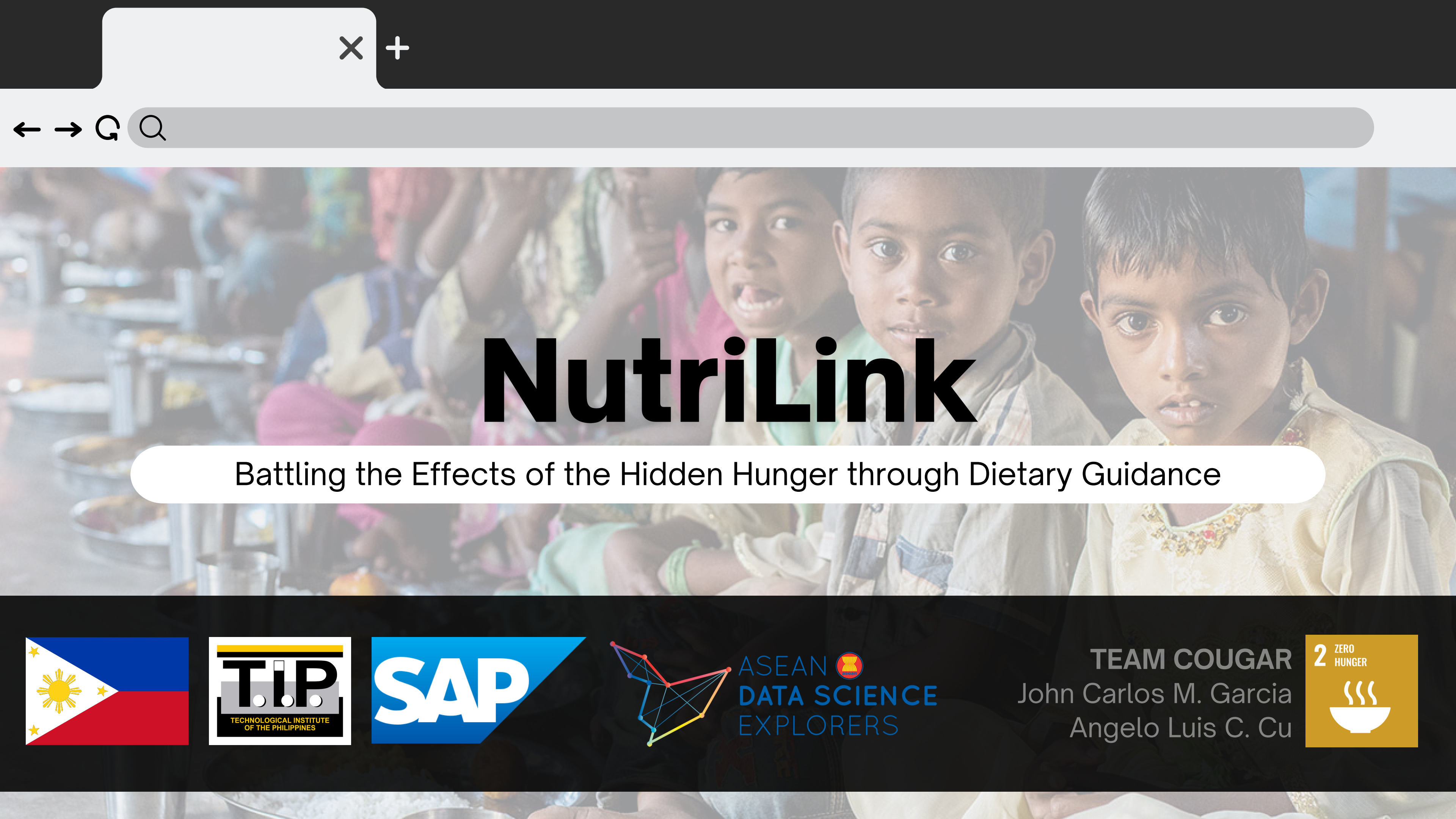




NutriLink

Battling the Effects of the Hidden Hunger through Dietary Guidance



ASEAN
DATA SCIENCE
EXPLORERS

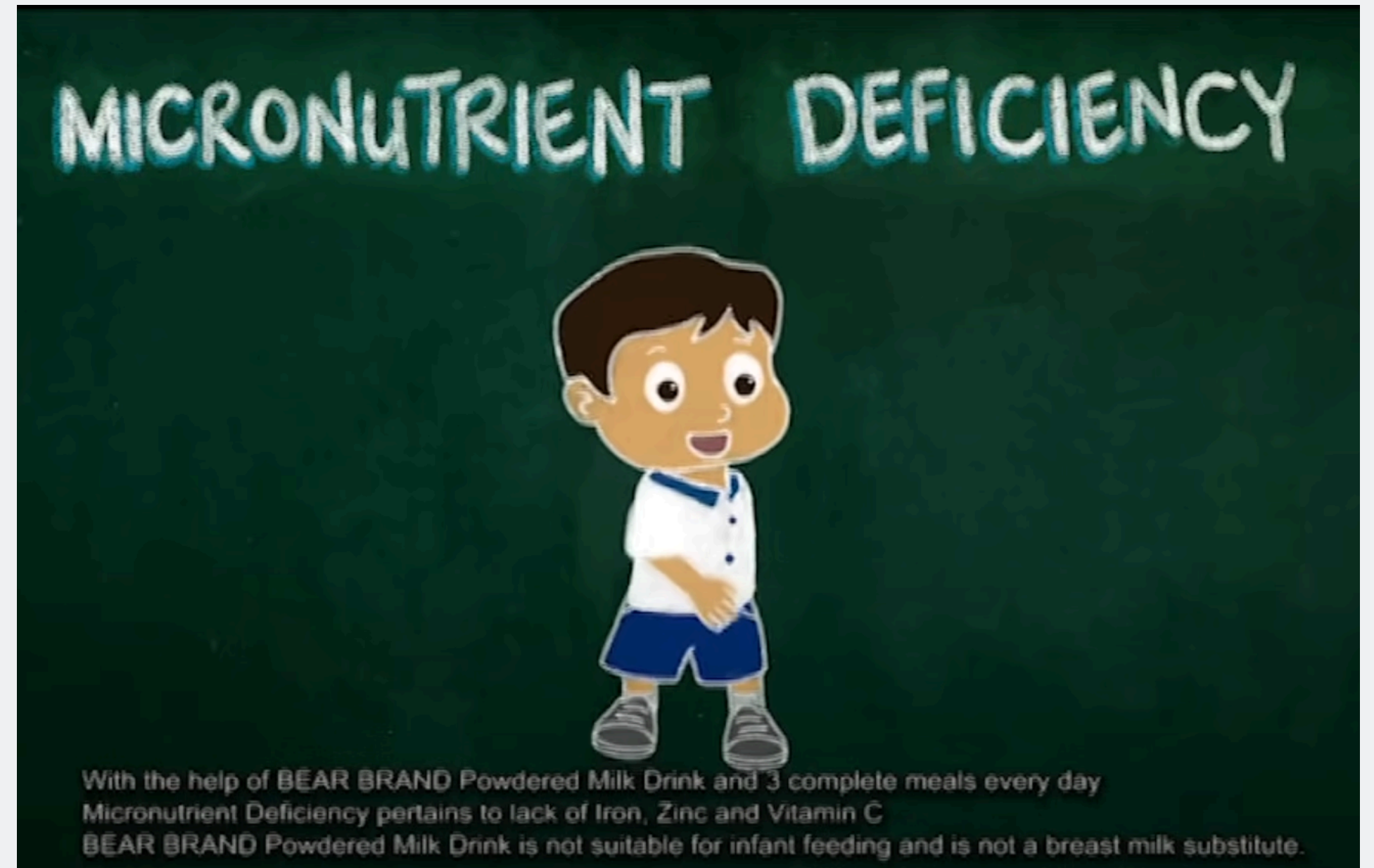
TEAM COUGAR
John Carlos M. Garcia
Angelo Luis C. Cu





← → 🔍 Introduction

Are you familiar with this advertisement by Nestle last 2015 in the Philippines which talks about a different kind of hunger?



Link: https://www.youtube.com/watch?v=E_ua03zGppc



← → 🔍 Brief Description

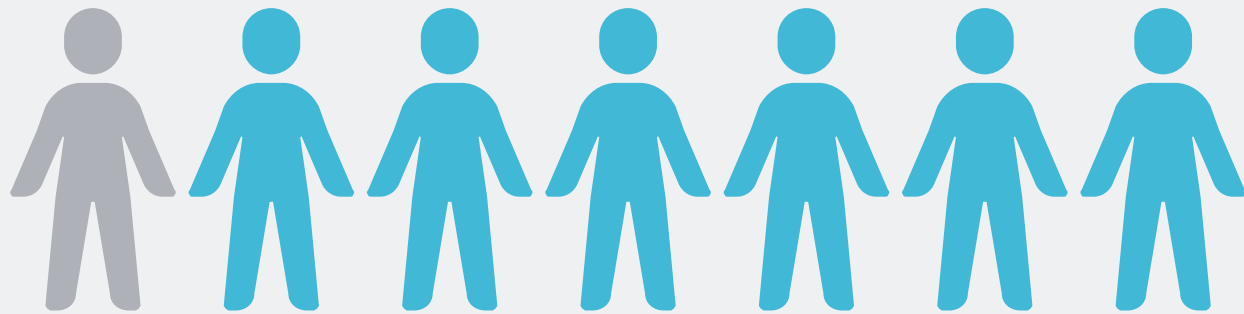
This is the **Hidden Hunger**

Also known as **micronutrient deficiency**, it is the lack of essential vitamins and minerals that are required in small amounts by the body for proper growth and development.

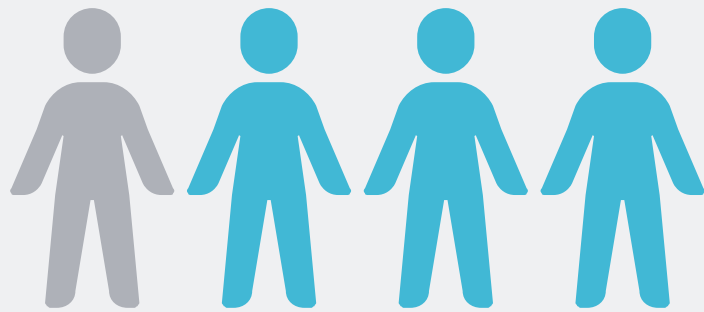
According to the World Health Organization (WHO)
and Our World in Data (OWID)

In the Philippines last 2019...

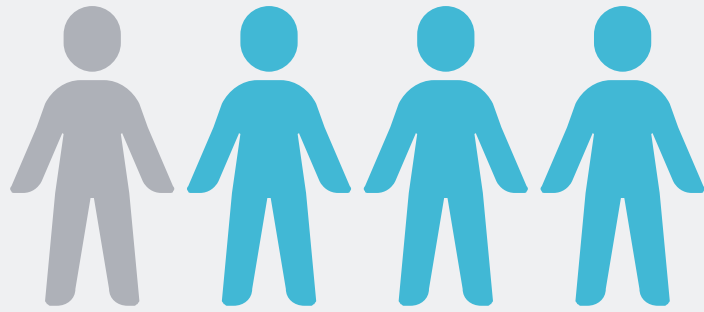
According to 2019 Expanded National Nutrition Survey, FNRI-DOST



1 out of 7 children has Vitamin A Deficiency



1 out of 4 children has Iron Deficiency



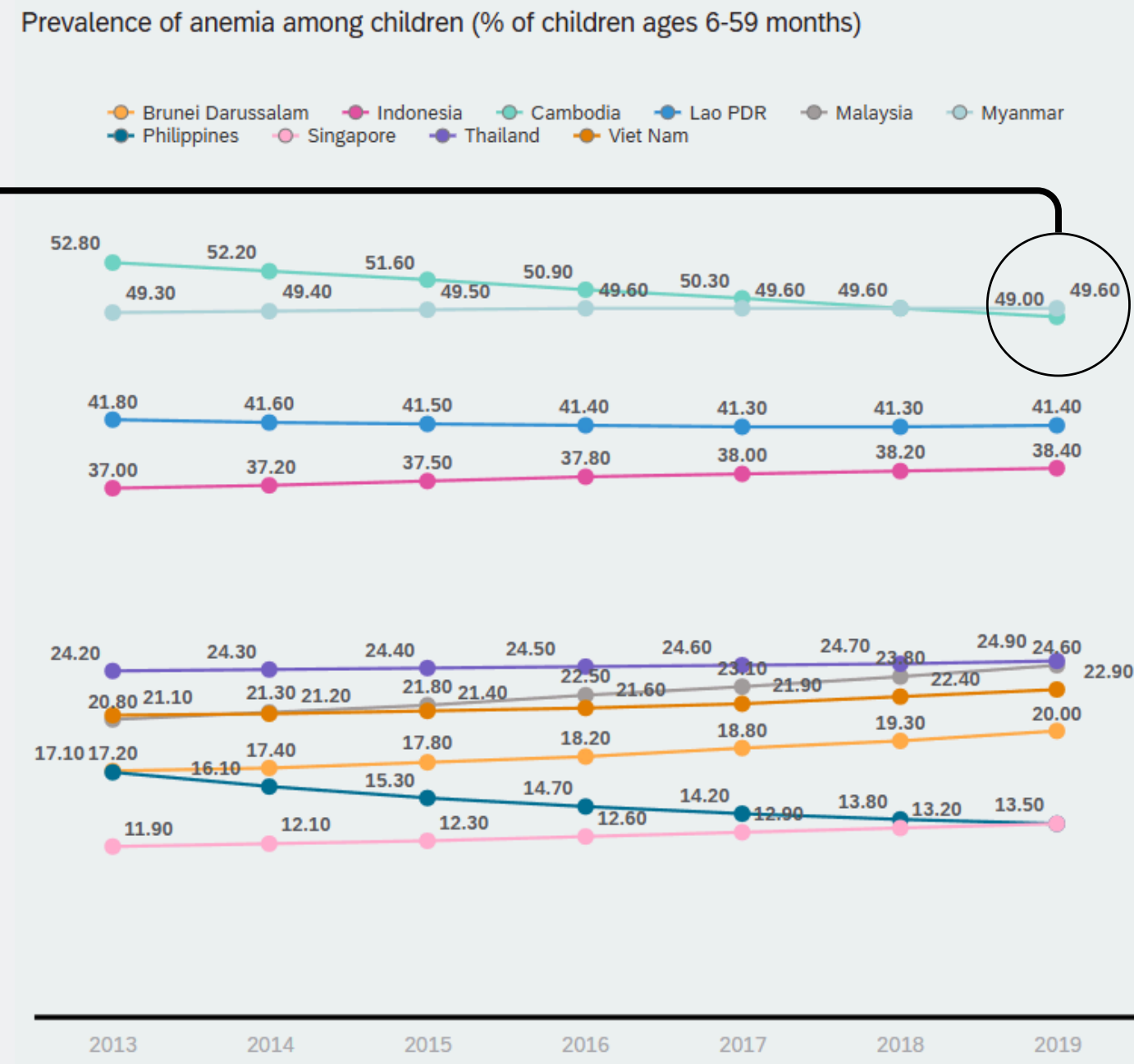
1 out of 4 children has Zinc Deficiency

Impacts of the Hidden Hunger

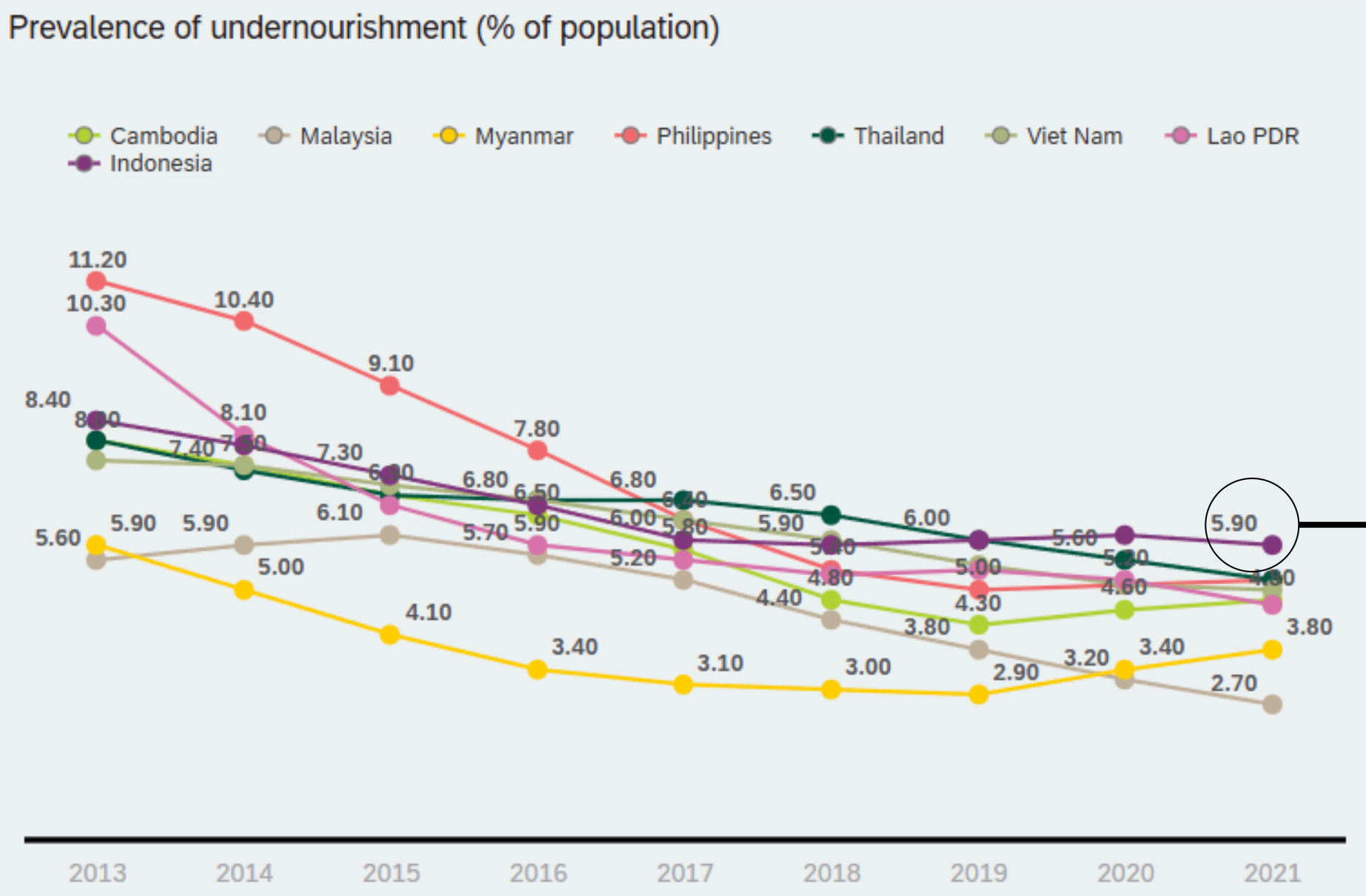
Children can develop Anemia

Myanmar, along with Cambodia, had the highest prevalence of anemia among children ages 5 and below in 2019

Data from World Bank Open Data



Impacts of the Hidden Hunger

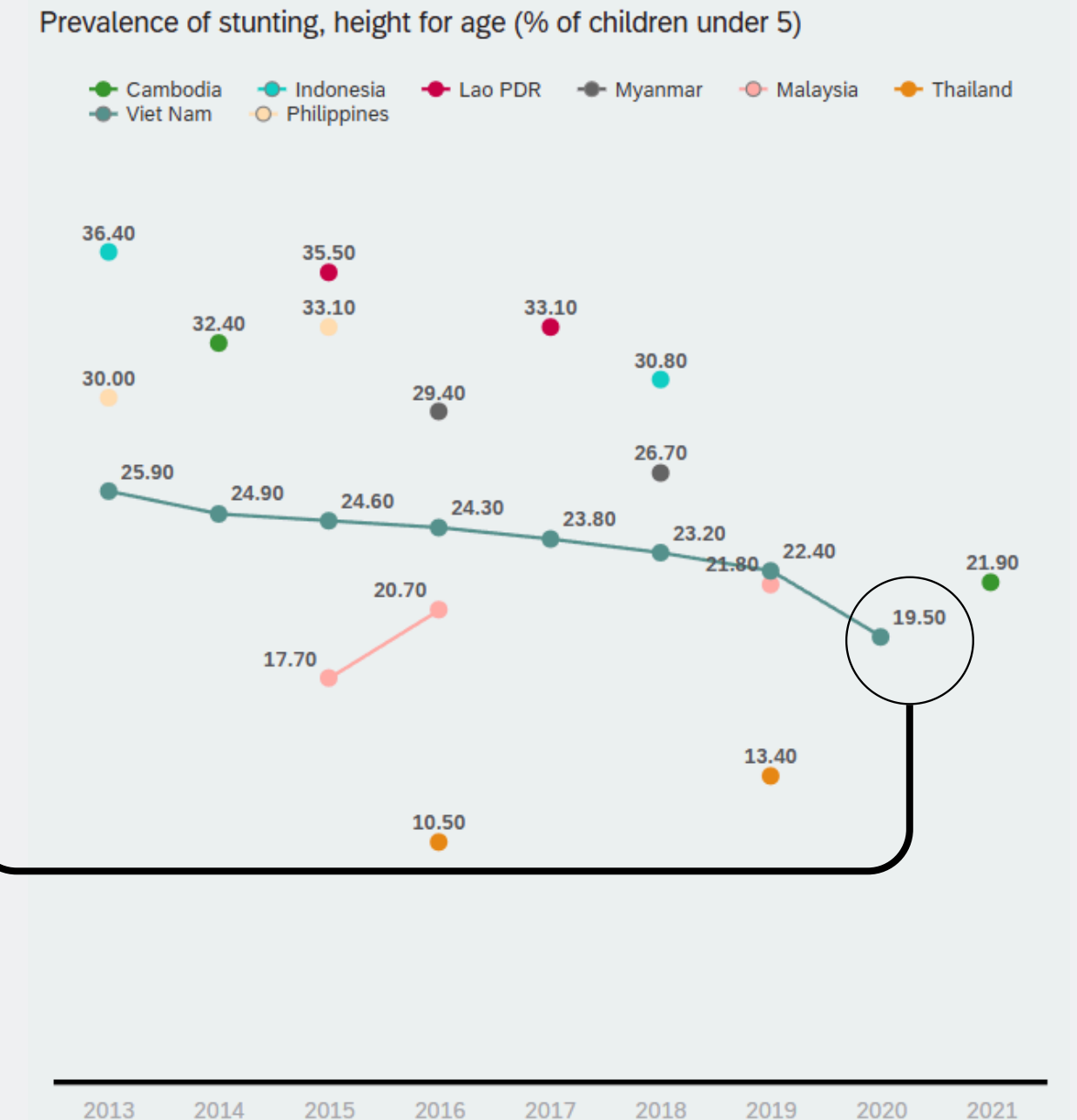


Undernourishment
Indonesia has the highest prevalence of undernourished people among ASEAN

Data from World Bank Open Data

Stunted Growth

Data from World Bank Open Data



All NewsEconomyMedia OutReachSocial MediaSocio-Economic Development

Skyrocketing Vegetable Prices Highlight Cost of Living Crisis in Laos

By Chono Lapuekou · August 16, 2023

13970

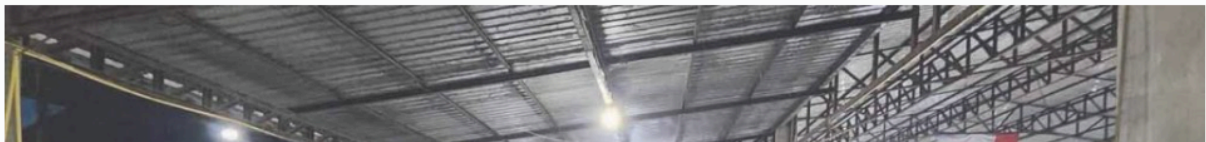
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Home > The Nation > 1 in 3 Filipino kids stunted — study

THE NATION

1 in 3 Filipino kids stunted — study

May 16, 2024 | 8:41 pm

Nearly 30% of kids in M'sia have stunted growth due to unhealthy diet, says Health Ministry

By BENJAMIN LEE, MARTIN CARVALHO, RAGANANTHINI VETHASALAM and JUNAID IBRAHIM

NATION

Wednesday, 01 Nov 2023

1:22 PM MYT

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Let us dig deeper, what underlies these current issues?

TheStar

Healthy diet cost in Brunei second highest after Indonesia in South-East Asia

Healthy diet cost in Brunei second highest after Indonesia in South-East Asia

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BRUNEI

Saturday, 25 Mar 2023

1:47 PM MYT

NEWS

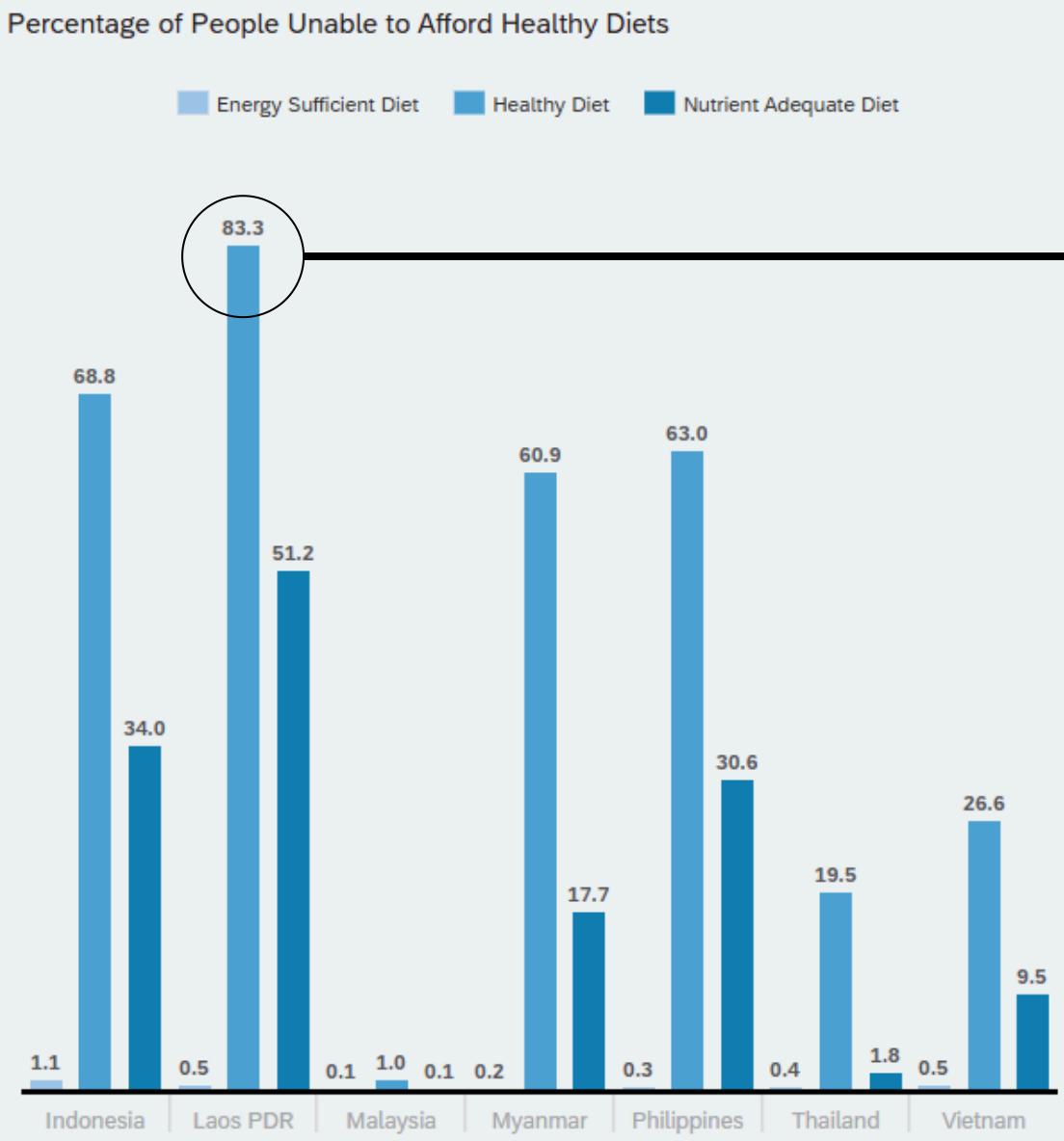
High price of healthy eating in Thailand impacts young people

4 May 2023

Consumers in Thailand say they want to eat healthier but that fresh, nutritious food is too expensive and inaccessible for the average person, according to Mintel.

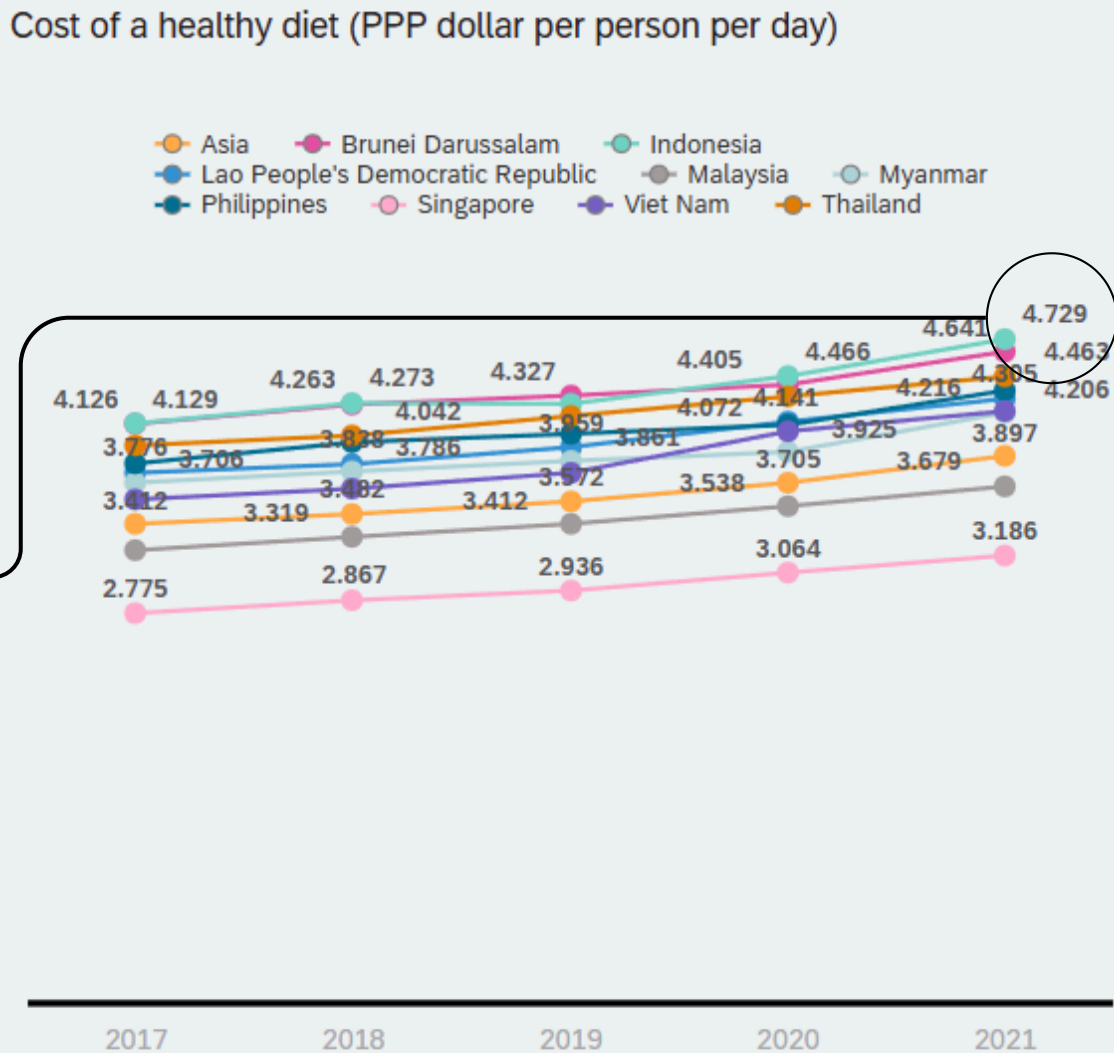
MONEY

According to Ritchie and Roser (2024), a healthy, nutritious diet is **much more expensive** than a calorie-sufficient one.



Laos has the highest percentage of people unable to afford healthy diets (83.30% in 2020)

Indonesia has the highest cost for a person's healthy diet for a day (4.729 dollars in 2021)



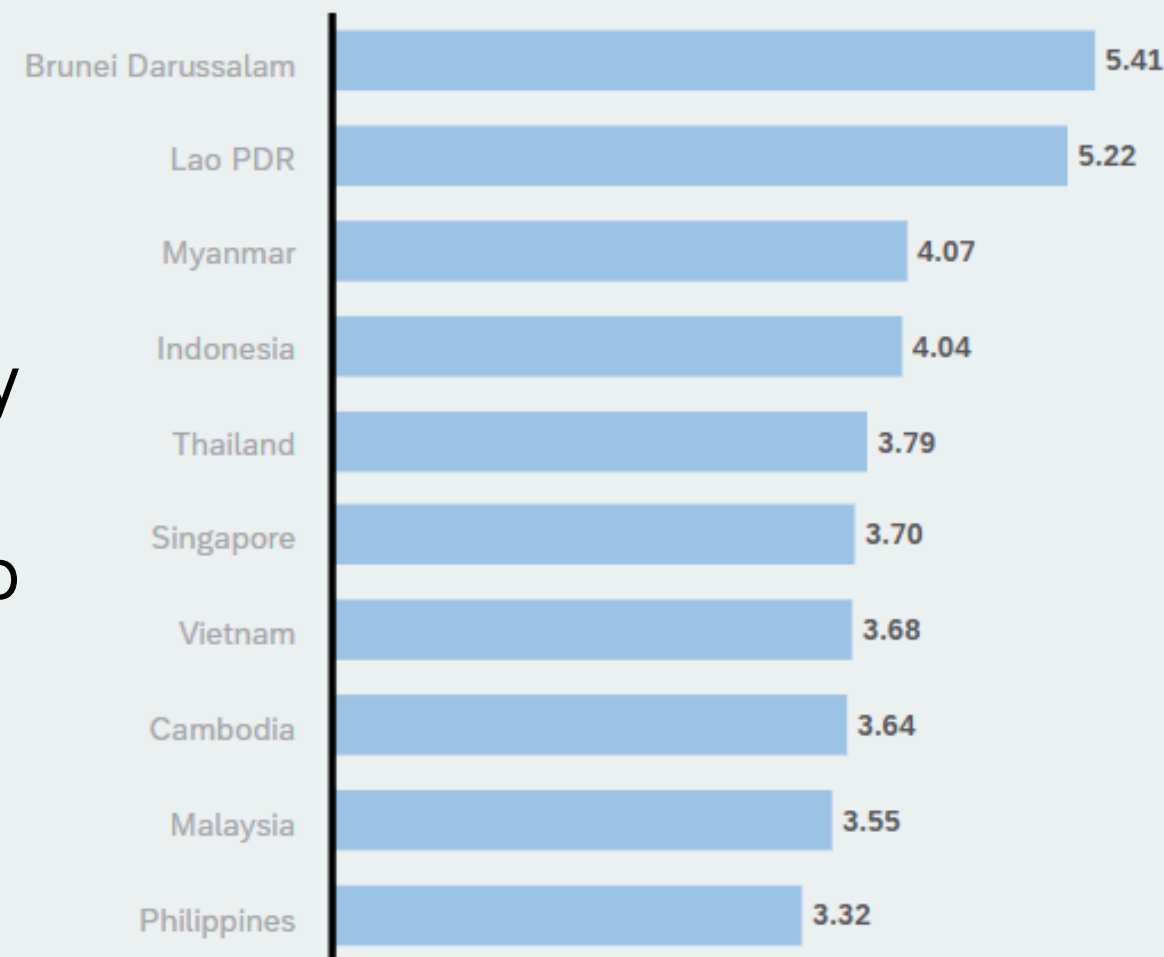
CULTURE

- Rice-heavy diet
- Shift from market to grocery shopping
- More convenient and cheap unhealthy foods

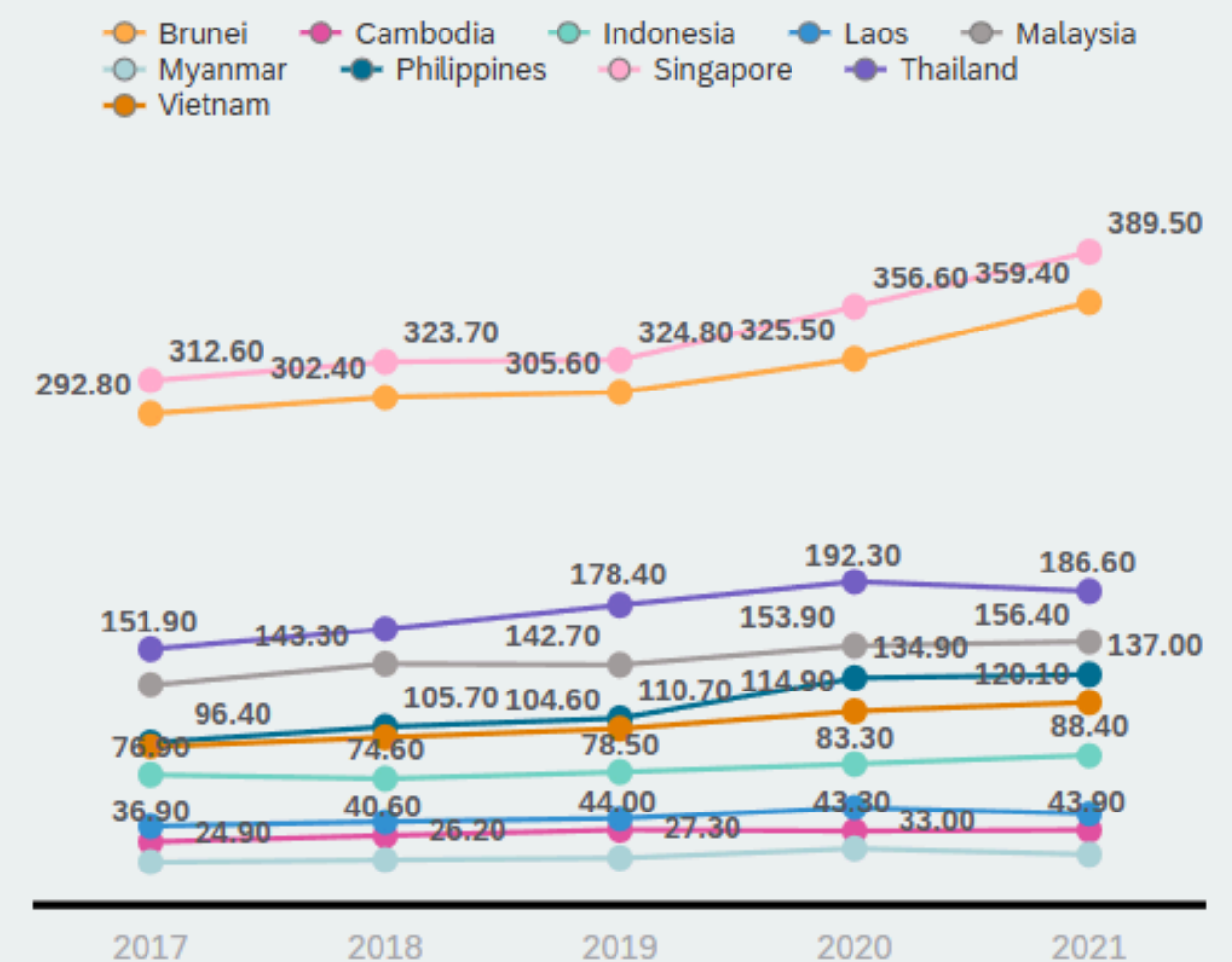
Led to a transition from **traditional and Indigenous diets** toward **processed foods**

Data from World Bank

Cost of a healthy diet relative to the cost of sufficient energy from starchy staples (Ratio, USD)



Total sales of ultra-processed foods per person (current (nominal) US\$/capita)



← → 🔍 Recommendation

NutriLink

Select Information

Name

Type here...

Age

Select option

Submit

Hello, <Name>

Food Suggestion for the day:

Proposed Solution

NutriLink

NutriLink is a mobile application that provides a personalized meal plan based on the user's micronutrient needs and budget.

Test it out!

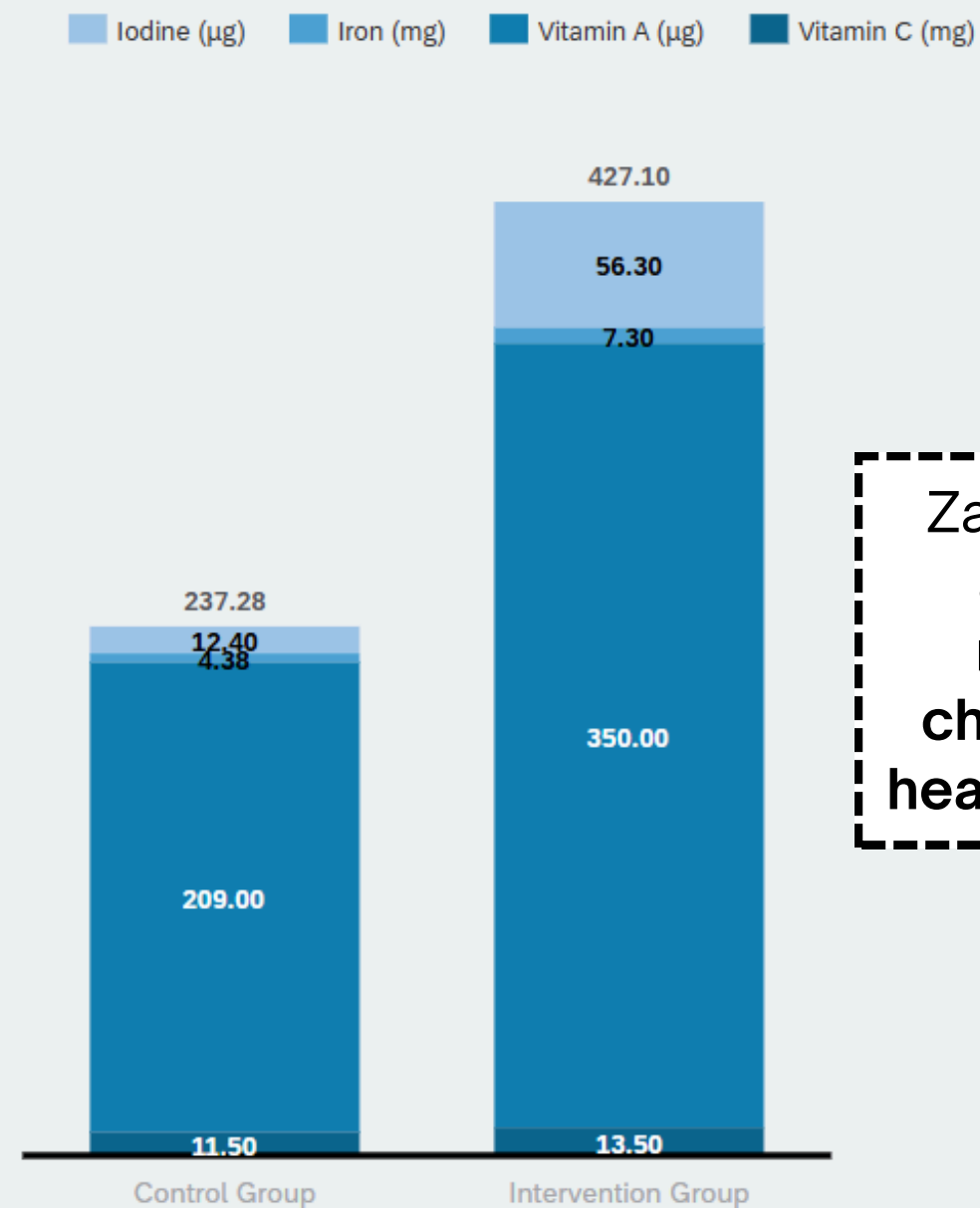
Demographic Data

NutriLink

Meal Plan

How feasible are mobile applications for Dietary Guidance?

Consumption of Micronutrients

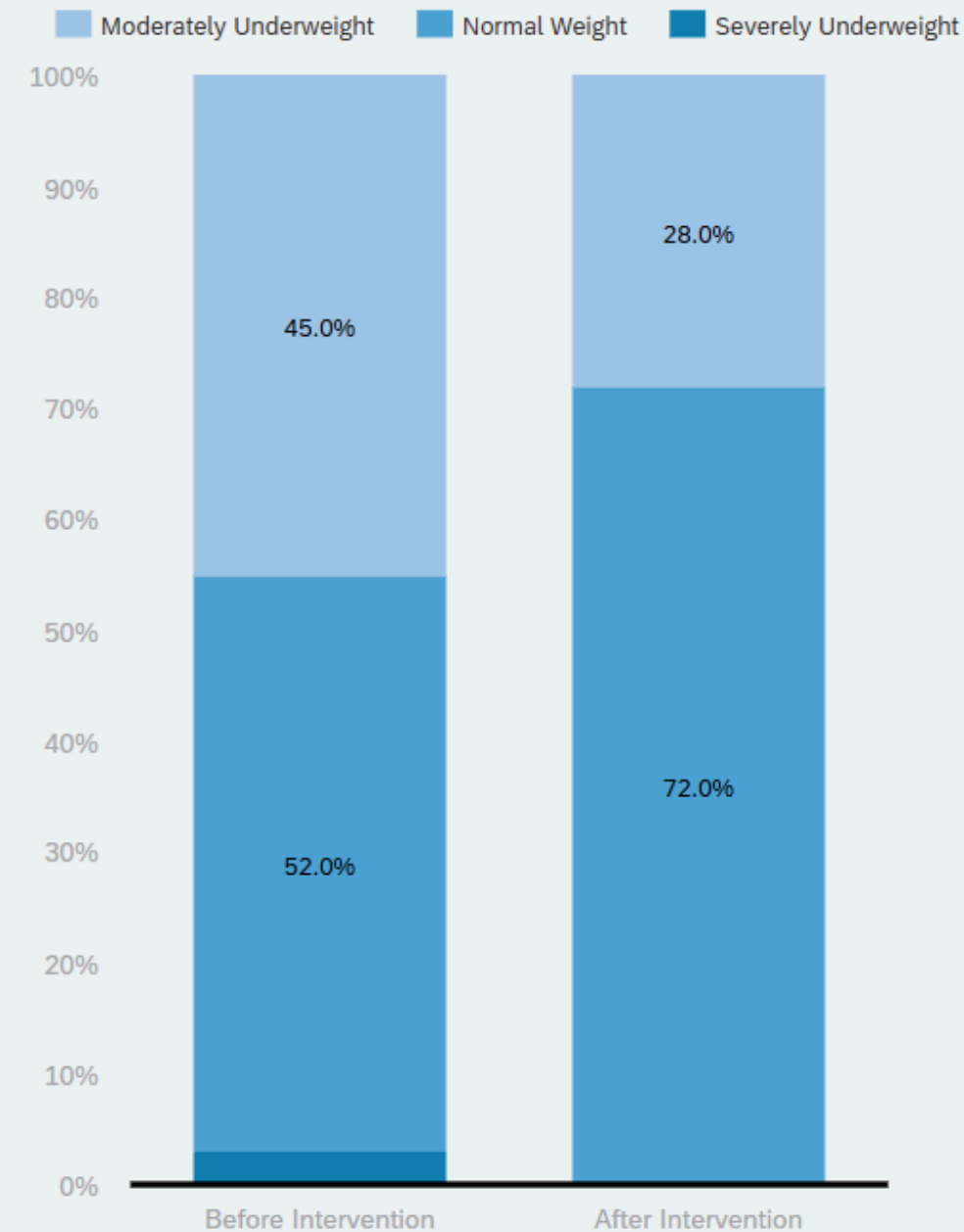


Zahid et al. (2023) were able to **improve the nutritional intake of children with the use of health-based mobile apps**

Data from Zahid et al. (2023)

Percentage of Children's BMI Before and After Intervention

1 Filter



West et al. (2017)
found that people
are **59.9% more**
likely to eat
healthier diets with
nutritional apps

NutriLink contributes to the attainment of SDG 2

2 ZERO HUNGER



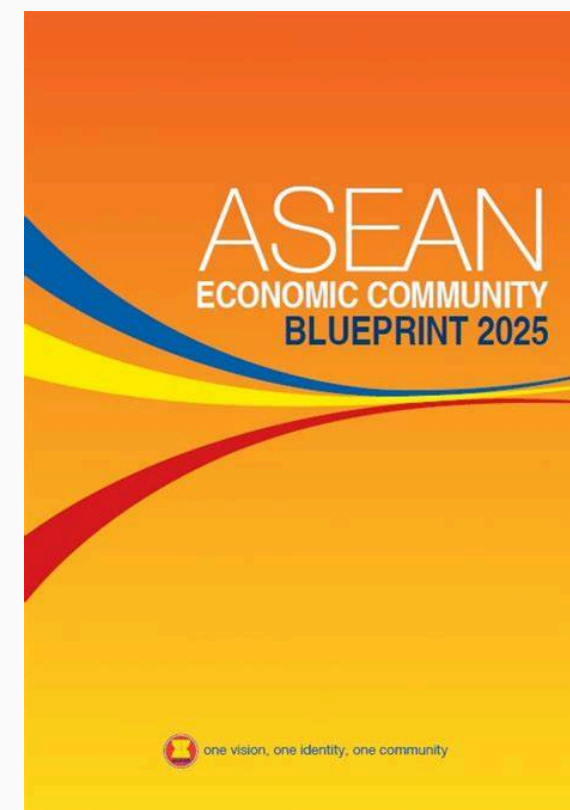
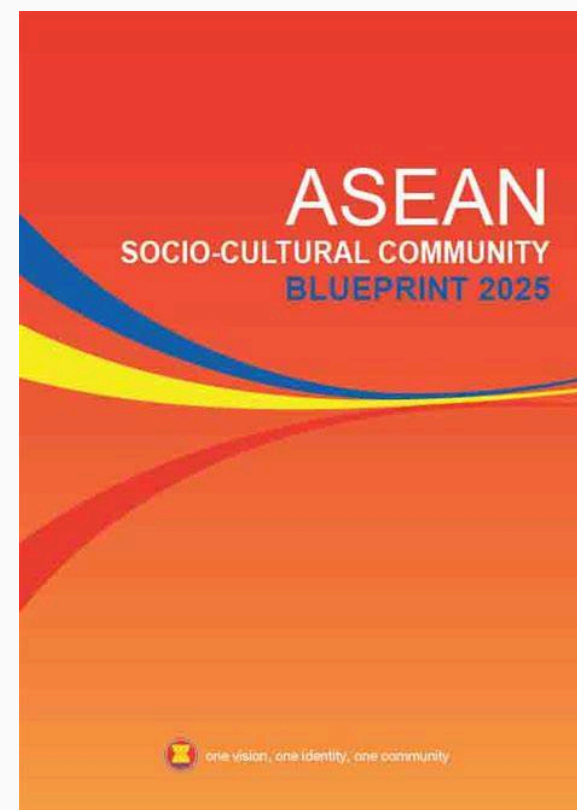
Focusing on **Goal 2.2.s1: Prevalence of micronutrient deficiencies**

“Food security requires a **multi-dimensional approach** – from social protection to safeguard safe and nutritious food especially for children to transforming food systems to achieve a more inclusive and sustainable world.” (un.org/sustainabledevelopment/hunger/)

ASEAN Blueprint for 2025

ASEAN Socio-Cultural Community Blueprint 2025

D.5.i: Enhance ability to cope with food price by developing adaptive mechanisms and strategies



ASEAN Economic Community Blueprint 2025

C.5.56: Contribute prosperity in the ASEAN community by ensuring food security, food safety, and better nutrition.

Project Roadmap

Year 1

Consultation with Licensed Nutritionists and National Food Security Councils



Years 5-6

App testing then deployment along with publicity campaigns



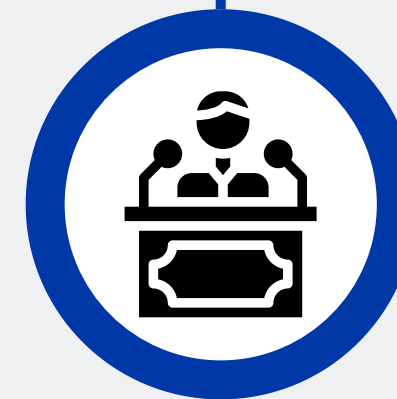
Years 2-4

App Development



Year 7 onward

Getting continuous user feedback for further improvements, along with maintenance





Thank You!

TEAM COUGAR

John Carlos M. Garcia

Angelo Luis C. Cu

← → ↺ 🔍 References

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